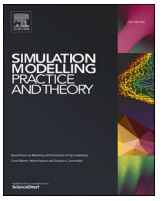
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Simulating optimal flood evacuation using heuristic algorithms and path-choice behaviors

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ABSTRACT

Effective path planning in flooding emergency rescue scenarios is essential for ensuring timely evacuation while minimizing safety risks. Conventional path-planning algorithms often prioritize the shortest or most cost-efficient routes, potentially neglecting safety considerations. To address this limitation, this study introduces an improved path-planning method using a behavior-based A-star (A*) algorithm designed for evacuation scenarios. A cellular automata (CA) environment is applied to address common challenges associated with traditional A* algorithms, including path inefficiencies, longer distances, and difficulties in handling dynamic flood environments. The key innovation of this study is the optimization of a heuristic function by integrating depth sensitivity perception (DSP), which directly influences evacuation behavior by prioritizing safer paths based on real-time water depth assessments during path selection. Experimental results across diverse flood scenarios demonstrate that the optimized A* algorithm significantly outperforms traditional A-star and Dijkstra's algorithms, achieving reductions in explored nodes by 90.06 % and 93.13 %, lowering safety risks, and shortening computational times by 87.65 % and 88.06 %, respectively. These findings validate the efficacy of the depth-sensitive heuristic in enhancing evacuation pathfinding within complex flood environments.

1. Introduction

Floods are among the most devastating natural disasters, inflicting social, economic, and environmental losses and posing substantial challenges to cities worldwide [1,2]. The combined effects of global warming and rapid urbanization have exacerbated flood risk factors in urban areas, underscoring the urgent need for resilient emergency response strategies [3]. Evacuation is an essential part of disaster management, involving the movement of people from flood-prone zones to safer areas [4]. Path modeling is a critical component of evacuation planning by identifying optimal routes and improving overall efficiency [5]. Nevertheless, risk management in evacuation planning faces uncertainty, necessitating hazard modeling solutions using simulation experiments [6].

Urban flooding presents significant challenges in rescue path planning, requiring individuals to navigate complex relationships to

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ensure safety [7]. Various models, particularly grid-based approaches, effectively represent urban scenarios and enhance path planning efficiency. The cellular automata (CA) model is especially useful for analyzing the spatiotemporal complexity of flood disasters [8]. While this model emphasizes rapid disaster response, it also encounters challenges due to uncertainties in immediate risks and obstacles [6]. Additionally, it can simulate human evacuation, providing critical data on water depth and recession [9]. However, previous research has not adequately addressed the internal behavioral parameters—such as depth risk perception, evacuation decision thresholds, and path choice preferences—in simulating human movement. Incorporating these parameters and behavioral analyses into an agent-based model (ABM) could significantly improve our understanding of human-environment interactions and enhance decision-making [10,11].

In evacuation planning, risk perception plays a crucial role, influencing the timing and decisions of evacuees [12]. Individuals subjectively evaluate risks when making evacuation choices, often prioritizing efficient routes while evaluating perceived hazards [13]. Although research has examined flood-related behavior, the relationship between internal decision-making and floodwater conditions remains unclear [14]. Current studies on rescue path planning mainly emphasize evacuation systems and flood resilience. Despite growing recognition of grid-based planning's importance, a significant research gap exists in addressing uncertain risk factors and the complexities of urban environments [15]. Furthermore, existing algorithms such as A* and Dijkstra's have not fully resolved these challenges [16].

Pathfinding algorithms, including best-first search, Dijkstra, and A* algorithms, are commonly used in navigation problems. Breadth-first search (BFS) is more efficient than Dijkstra but does not always guarantee the optimal path [17]. Dijkstra's algorithm ensures optimality but is more computationally complex [18]. The A* algorithm, a heuristic search algorithm, has been improved for various applications, such as optimizing heuristic functions for smoother paths and improving cost functions for unmanned aerial vehicles (UAVs) [19–21]. However, most A* variants predominantly prioritize distance [22], a limitation particularly evident in dynamic environments like flood rescues, where the shortest path may not be the safest.

The central motivation of this research is to integrate behavioral dynamics into rescue models, emphasizing evacuees' risk perception and decision-making processes. Accordingly, this study aims to develop an optimal A* algorithm for flood rescue planning that incorporates depth sensitivity and behavioral factors into pathfinding. Moreover, the research enhances the algorithm's heuristic function to minimize safety costs and improve evacuation efficiency. The proposed approach is validated by simulations in CA and a realistic environment, comparing its performance with traditional pathfinding algorithms. Section 2 presents a comprehensive literature review, while Sections 3, 4, and 5 detail the methodology, experimental simulations, results, and discussion. Lastly, Section 6 concludes the study and proposes directions for future research.

2. Literature review

Recent studies have highlighted the necessity for flood-adaptive optimal path strategies [23]. However, determining safe evacuation approaches remains challenging due to the influence of risk perception and human factors such as mobility, age, and gender [24]. Furthermore, effective flood rescue plans must account for the dynamic nature of flooding events [25]. Several modeling approaches have been developed to simulate flood evacuation processes, including hydrodynamic models such as two-dimensional (2D) simulations, which are widely used for flood impact mapping and risk assessment [26,27]. Lumbroso and Davison [28] introduced the life safety model to evaluate human safety during flood events through hydrodynamic simulations. Similarly, Dawson et al. [29] and Liu & Lim [30] employed empirical and hydrodynamic methods to predict evacuation outcomes. Despite these advancements, existing approaches have primarily focused on scenario-based analyses, often overlooking the optimization of evacuation paths under dynamic flood conditions.

Water depth significantly impacts flood rescue safety and risk perception, with identified thresholds helping in developing efficient evacuation plans [31]. Eduardo et al. [32] established safety thresholds for children and adults based on water depth and velocity, highlighting their importance in evacuation planning. Zhang et al. [33] utilized geographic information systems (GIS) to analyze impassable paths under flood conditions. Hu et al. [34] incorporated water depth thresholds in shelter delineation and evacuation path planning. Musolino et al. [35] introduced a mechanics-driven approach that considers human body attributes to improve evacuation efficiency. He et al. [36] emphasized integrating hydraulic variables (e.g., water depth) with simulation tools developed using ABM and CA. This combination creates a comprehensive framework for analyzing evacuation planning to bridge the gap between physical risk assessment and human behavior modeling in dynamic flood scenarios. ABM considers individuals as adaptive decision-makers, providing an understanding of human behavior during floods [29,37]. Johansson et al. [38] developed a social force model to capture psychological and physical responses during evacuations. Meanwhile, CA applies grid-based models to simulate dynamic environments efficiently [39,40]. Lim et al. [41] highlighted that integrating ABM and CA enhances flood evacuation planning.

Recent research has focused on advanced methodologies to improve flood evacuation strategies through integrated modeling tools [6]. Algorithms, such as A*, genetic, and probabilistic methods, have been applied to determine optimal evacuation paths [42]. Li et al. [43] improved evacuation algorithms by incorporating human instability heuristics. Blatt and Szczerbicka [44] explored multi-agent systems for flood-prone areas. Dubois et al. [45] defined path planning as a capacitated vehicle routing problem, and He et al. [46] proposed a dynamic optimization approach that considers flood dynamics and individual response times. Numerous studies have overlooked dynamic hydraulic variables such as water depth.

The A* algorithm is effective in path planning, leveraging heuristics and cost functions to identify optimal routes. Bernardini et al. [47] showed its efficacy in emergency evacuation planning, while Dawson et al. [29] expanded its applications by incorporating hydraulic factors in flood-specific modeling. Du et al. [48] improved the algorithm by incorporating social and environmental considerations. Zheng et al. [49] applied it to optimize evacuation paths in underground metro stations and demonstrated its suitability for

constrained environments. Ozcan et al. [42] pioneered the use of A* algorithm for rescue boat planning in a flooded urban environment using UAV-derived aerial imagery.

Despite these advancements, the flooded urban environment remains unexamined with the proposed approach. Therefore, further

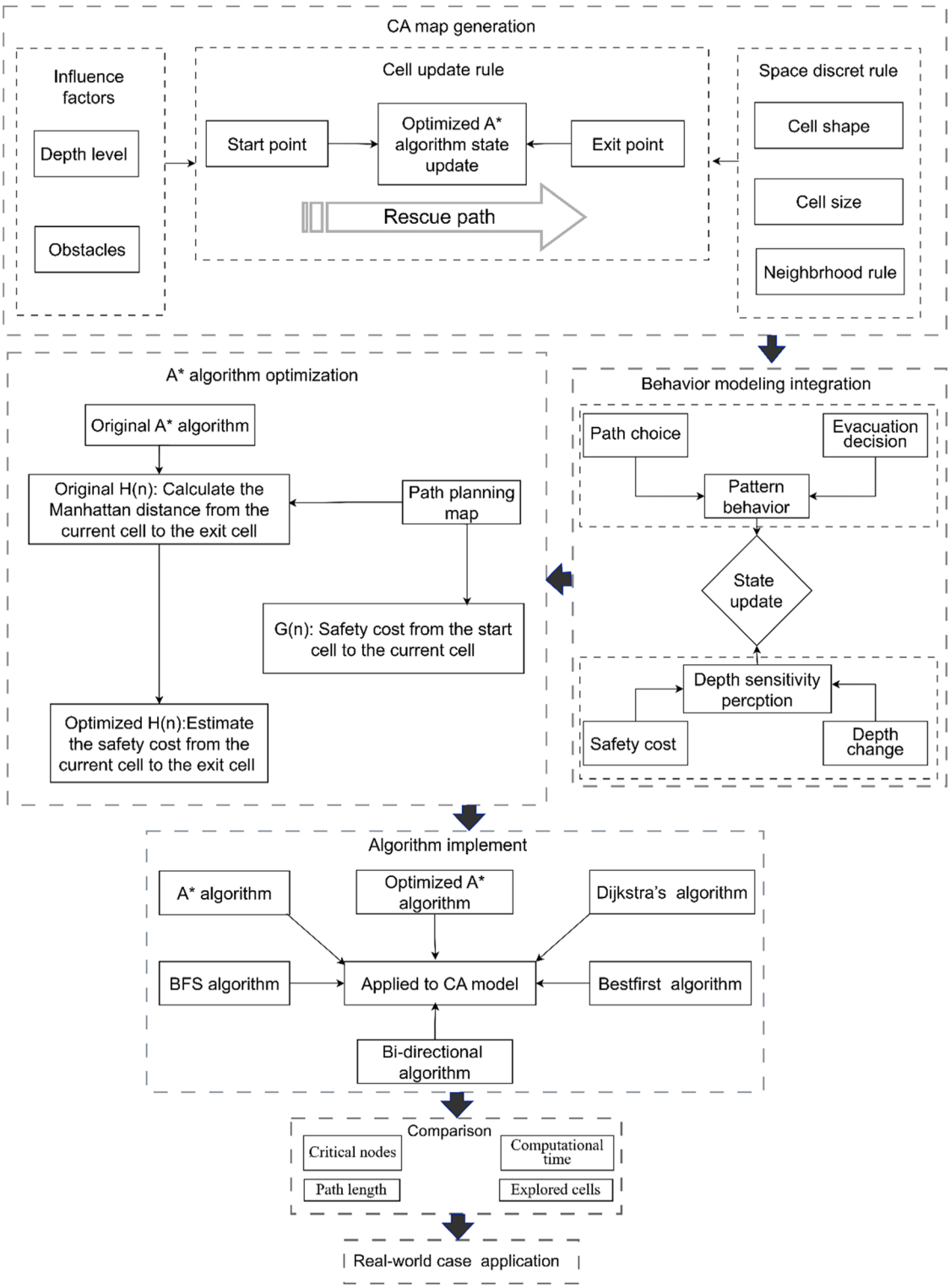


Fig. 1. Methodological framework.

studies are needed to improve the performance of path planning in complex scenarios combining behavior modeling and dynamic flood environments. An optimized algorithm that captures individual behaviors to find the optimal evacuation path in areas with varying flood depths is proposed. This study addresses a notable gap in existing research by incorporating behavioral patterns as constraints within the optimization process. Previous studies have tended to underestimate or overestimate parameters by neglecting the dynamic aspects of evacuation behavior [50]. In contrast, our approach integrates two key behavioral factors: evacuation decision-making and path choice. This work also introduces a novel focus on depth sensitivity perception (DSP), modeled as an internal state that directly influences agent behavior.

3. Methodology

3.1. Overview

Fig. 1 illustrates the technical framework of this study, beginning with the CA map generation, which integrates depth level and obstacle data through spatial discretization rules, including cell shape, size, and neighborhood configuration, alongside a cell update rule. Subsequently, behavioral modeling is integrated to dynamically adjust evacuation decisions and path choices based on depth sensitivity and behavioral patterns. The A* algorithm is then optimized by refining its heuristic function to estimate safety costs, leveraging the Manhattan distance and safety costs as key metrics for improved path planning. Next, the framework applies the optimized A* algorithm alongside five path planning algorithms within the CA model, comparing performance across four parameters: computational time, path length, explored cells, and critical nodes. Finally, the optimized A* is implemented in realistic cases with multiple scenarios to assess its practical applicability.

3.2. Cellular automata

The proposed evacuation simulation in the CA environment can be formally expressed as a quadruple [51], where each cell represents a 10 m × 10 m area updated according to predefined rules. CA can be formally defined as follows:

$$A = (Z^d, X, S, \sigma \tag{1}$$

where Z^d represents a lattice of cells covering the simulation grid, X is the definition of the local neighborhood, S is the set of cell states, and σ is the transition rule determining the changes in cell properties.

The transition rule is a stochastic function that introduces non-determinism into the model through probabilities, simulating unpredictable situations [41]. In alignment with this research, the model comprises four key components: a Moore neighborhood with quadrilateral cells in 2D space; cellular state divided into four parts, serving as the agent's starting point; cells designated as a safety point; flooded cells with varying water depths that remain static during simulation; and cells with fixed physical obstacles that cannot be occupied, all of which are relevant to evacuation path planning. The evolution rules emphasize that agents prioritize the safest path toward the nearest exit, allow multiple agents to occupy a single cell simultaneously, and enforce agent immobility if traversing a cell with water depth exceeding a predefined threshold during evacuation.

Fig. 2 demonstrates how the CA environment simulates dynamic water depths and obstacles, guiding agent movement decisions based on safety costs during evacuation path planning. Agents navigate the CA grid from cell C (i, j) by perceiving depth-related risks, avoiding high-risk cells with greater depths. At each step, agents evaluate all eight surrounding cells using a Moore neighborhood configuration to determine the safest path. These neighboring cells include both orthogonal and diagonal directions (e.g., C (i ± 1, j), C (i, j ± 1), and C (i ± 1, j ± 1)). The deterministic transition function updates cell states based on predefined conditions, depth levels, and obstacle configurations, requiring agents to employ depth-sensitive perception for navigation. Additional insights into agent-based

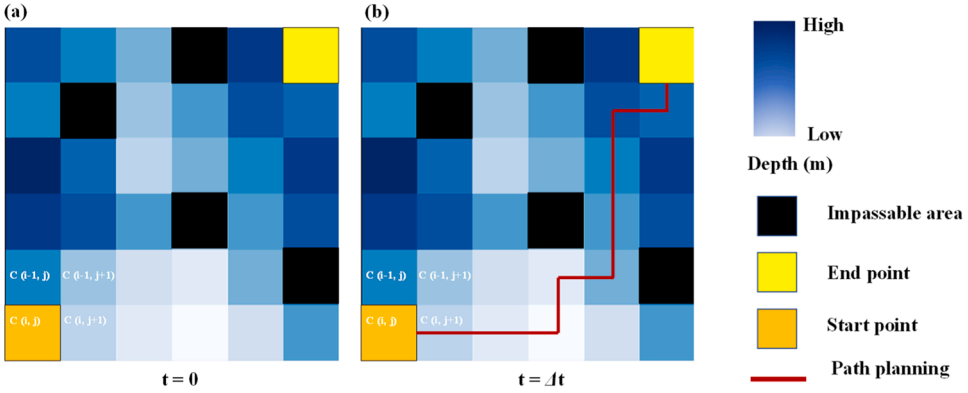


Fig. 2. Schematic diagram of CA environment: (a) initial position and state of the grid at t = 0, and (b) updated agent path planning change at t = Δt.

behavioral integration and the path planning algorithm are elaborated in Sections 3.3 and 3.4, respectively.

3.3. Agent-based model

ABM simulates human behavior in a flood environment characterized by varying water depths. The subsequent sections describe the structure, patterns, and behavior modeling based on depth risk sensitivity.

3.3.1. ABM structure

In ABM, each agent represents a pedestrian navigating a flood-affected area with varying water depths and obstacles. ABM captures the diversity of agent behaviors, focusing on assessing environmental risks, distance to exits, and physical barriers. Agents are reactive to their surroundings and guided by an internal state (DSP) that influences their evacuation decisions and path choices. ABM works alongside a CA environment, where agents evaluate neighboring cells based on depth levels and spatial constraints, ensuring that their responses reflect safety and distance trade-offs. The proposed ABM structure is shown in Fig. 3.

3.3.2. Behavior patterns

In the proposed ABM, agents exhibit behaviors during evacuation driven by their perception of safety cost and proximity to exits. However, these perceptions are based on real water depth data and are moderated by the agent's internal risk sensitivity threshold. Key behaviors are defined as two key actions: evacuation decision and path choice. Using CA rules, agents evaluate their surroundings

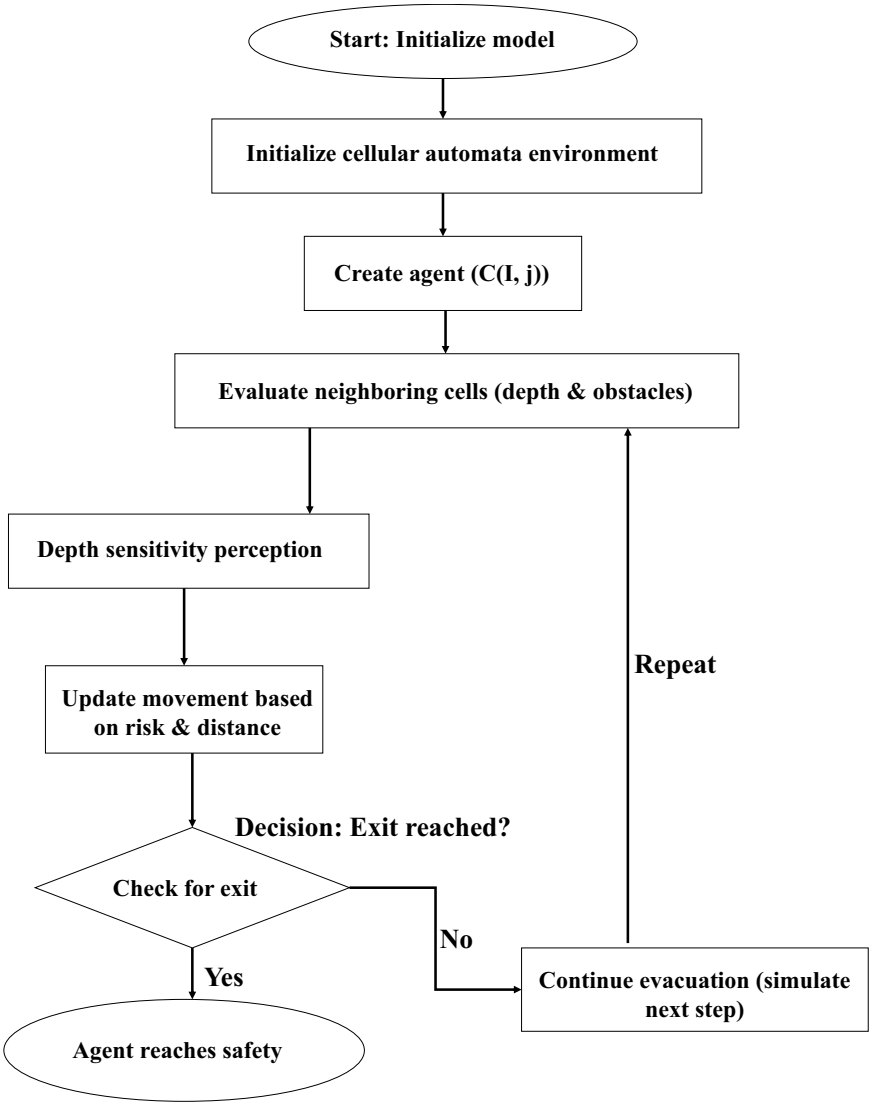


Fig. 3. ABM flowchart for evacuation.

throughout the evacuation decision phase, considering flood depth and obstacles. They decide to evacuate when the flood depth exceeds their internal sensitivity threshold, prioritizing safety as flood risks increase. Once the decision to evacuate is made, agents seek optimal paths to safety by dynamically assessing surrounding conditions.

In the path choice phase, agents effectively evaluate evacuation paths based on two main criteria:

- Safety cost: Agents prefer paths through shallower water to minimize hazard exposure.
- Distance to exit: Agents prioritize the shortest possible paths to evacuation exits.

This balance ensures dynamic decision-making, with agents continuously re-evaluating and adapting their paths as flood conditions change throughout the evacuation process.

The behavior pattern described is further illustrated in Fig. 4, which represents how agents in flooded scenarios assess environmental factors. Specifically, they consider the water depth level and distance to the exit. These factors influence their internal decision-making processes, guiding whether agents initiate evacuation immediately or reassess available paths. During path selection, agents dynamically balance safety cost and the shortest path, constantly adjusting routes based on updated risk evaluations to prioritize safety and efficiency. To streamline and standardize agent modeling, this study simplifies the inundation-agent interaction framework based on Hsu et al. [52] by standardizing agent characteristics using a single human instability index and adopting a uniform relationship between walking speed and water depth.

3.3.3. Behavior modeling-based DSP

This paper presents DSP as an internal state influencing agent behavior in a CA model. DSP significantly impacts agents' movement characteristics, evacuation decisions, and path choice, balancing shorter and safer paths. An optimization algorithm for the heuristic function in the A* algorithm is proposed in Section 3.4.2, focusing on safety costs related to DSP. This perception-driven mechanism is further improved by integrating the perceivable boundary (PB), illustrated in Fig. 5, which highlights the agent's visual field and determines the horizontal space in which the agent assesses environmental factors [53]. The PB dynamically updates with the agent's movements, offering an adaptive spatial context that corresponds with DSP by allowing the agent to identify high-risk areas and assess possible paths.

DSP of agent i at time step t is denoted by DSP i, t, updated based on water depth d (i, j) at the agent's current cell C (i, j):

$$DSP_{i,t} = \frac{d(i,j)}{D_{max}} \tag{2}$$

where D_{max} is the maximum possible depth in the environment. $DSP_{i,t} \in [0, 1]$, where 1 indicates the maximum depth risk perception. Each agent has a depth perception threshold θ_i. If the perceived depth $DSP_{i,t}$ exceeds this threshold, the agent will consider the path high risk to continue and will attempt to find a safer path if $DSP_{i,t} > \theta\ i$, then the agent changes path or stops. Agents prioritize neighboring cells with lower $d(i,j)$ values, ensuring safer paths toward the exit cell. The agent evaluates eight neighboring cells, selecting the cell with the lowest DSP value that meets its threshold. If $d(i.j)$ D_{max} critical, where D_{max} critical represents the critical water depth for drowning, the agent is considered immobilized. When all adjacent cell exceed the agent's depth perception threshold, the agent ceases movement and initiates a reassessment protocol. If no viable paths are identified within updated cell evaluations, the agent remains immobilized or adjust its threshold in accordance with predefined behavioral rules. In the current model, the agent remains stationary until a path within its acceptable safety range is determined.

Extensive research has been conducted to determine the critical thresholds of water depth that affect the agent mobility during flooding events [54–58]. This study synthesizes previous findings to examine the relationship between the evacuee walking speed and

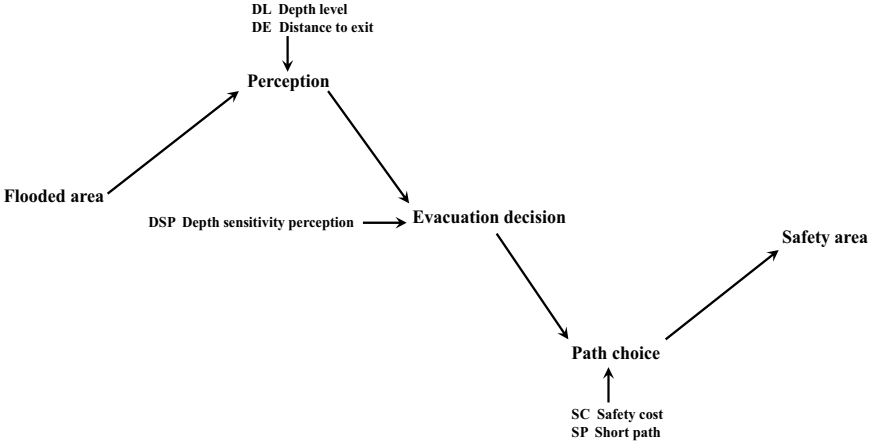


Fig. 4. Conceptual framework of the behavioral process for evacuation path planning.

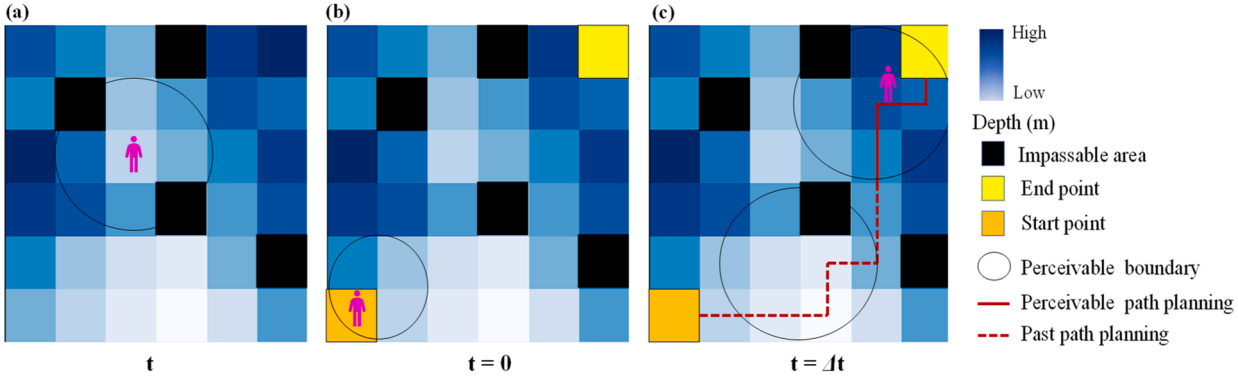


Fig. 5. Path planning evolution based on the perceivable boundary: (a) agent perception at time t, (b) agent initialization at t = 0, and (c) dynamic path adjustment at t = Δt as the agent moves.

the floodwater depth in ABM. Accordingly, critical water depth levels are categorized based on their impact on walking speed. At shallow depths (d ≤ 0.2 m), walking speed remains high (1.2–1.4 m/s), allowing agents to move with minimal disruption. Water depth increases to moderate levels ($0.2 < d \leq 0.5$ m), walking speed declines steadily to approximately 0.8 m/s, reflecting increased resistance, at higher depths ($0.5 < d \leq 0.9$ m), walking speed reduces significantly, dropping to 0.3–0.6 m/s, signaling considerable difficulty in movement. Finally, at critical depths ($d > 0.9$ m), agents are unable to proceed, as the walking speed reaches 0 m/s, representing a zone of immobilization and extreme risk (Hao-Ming [59]). This study considers $D_{max} = 0.9$ m and D critical = 0.6 m, with the critical depth representing a threshold to be avoided. The model explicitly addresses flooding scenarios involving stagnant or slow-moving water from prolonged rainfall rather than rapid flash floods, allowing a focus on depth-driven evacuation behavior [14]. This approach aligns with findings that risk perception escalates with water depth, streamlining computational efficiency while maintaining behavioral accuracy [60,61].

3.4. Path planning algorithm

3.4.1. CA construction

Before path planning, a CA construction grid must be established, as detailed in Section 3.2. Each cell within the grid is assigned a unique (x and y) coordinate to facilitate precise positioning, distance measurement, and pathfinding. The quadruple grid structure enhances agents' navigation efficiently while enabling accurate Manhattan distance computations, which are essential for optimal route determination.

3.4.2. Path search

The study employs the A* algorithm for path search, integrating principles from Dijkstra's algorithm and BFS [62]. The heuristic cost in the A* algorithm is derived from the distance between the current point and the endpoint [20], as expressed in Formula (3):

$$f(n) = g(n) + h(n) \tag{3}$$

where g(n) denotes the minimum cost path from the start point to the current node, while $h(n)$ estimates the optimal path cost from the current point to the exit (Hart et al., 1968). Conventional A* implementations rely on geometric heuristics such as Manhattan or Euclidean distance [63,64] which neglect behavioral factors critical to evacuation scenarios, such as perceived flood risk. To address this limitation, our study reformulates the heuristic calculation as a binary optimization problem, prioritizing DSP over simple geometric distance. This adaptation ensures alignment with agent-based behavioral modeling, where safety considerations drive evacuation decisions.

The heuristic function $h(n)$ based on DSP is adjusted, incorporating $d(i,j)$ at each cell and integrating a safety cost into the original distance calculation. The algorithm function is shown in Formula (4):

$$g(n) = \sum_{i=1}^{n} Si \tag{4}$$

where Si represents the safety cost of traversing cell i. Safety cost Si is based on $d(i,j)$ in cell C (i, j), embedding DSP into path planning to prioritize risk-averse routes. In Formula (2), if the weight of $h(n)$ is significantly larger than $g(n)$, the evaluation function $f(n)$ is mainly influenced by $h(n)$, causing the A* algorithm to be like a BFS algorithm, prioritizing heuristic estimation over path optimality [16]. Conversely, if $g(n)$ outweighs $h(n)$, computational efficiency is sacrificed for path optimality [65]; [66]. To optimize search efficiency, the $h(n)$ is refined as demonstrated in Formulas (5)-(8):

$$h(n) = S_{min} \times l_n \tag{5}$$

$$Si = w_d \times Depth(i,j) \tag{6}$$

$$S_{min} = \min\,(CS_k), \forall k \ \in untraversed\ cells \tag{7}$$

$$l_n = \ |\ x_n - x_e\ |\ +\ |y_n - y_e\ | \tag{8}$$

Where:

- w_d denotes the depth influence weight, aligned with DSP modeling,
- S_{min} guides agents toward the lowest-risk direction,
- l_n is the Manhattan distance, ensuring computational efficiency.

While agents assess all eight neighboring cells during navigation, the heuristic function's distance component is computed using Manhattan distance. This approach simplifies calculations and aligns with the grid-constrained movement logic inherent to the CA environment. Unlike Euclidean distance, which requires computationally intensive square root operations, Manhattan distance relies solely on basic arithmetic—addition, subtraction, and absolute values—enhancing computational efficiency [67]. This efficiency is essential in repetitive pathfinding calculations. Although Manhattan distance inherently restricts movement to orthogonal directions, this geometric limitation is mitigated by agent behavior rules driven by DSP. Hazardous cells are penalized according to water depth, ensuring agents avoid high-risk paths, even when diagonal alternatives exist. As demonstrated in Fig. 6, Manhattan distance effectively quantifies grid-based step counts between points, providing a computationally efficient yet behaviorally informed foundation for A* pathfinding.

4. Experiments and results

4.1. Experimental setting

A CA model was implemented in Python, leveraging an Intel i7–12700KF CPU and an NVIDIA RTX 3060 GPU to simulate flood scenarios using grid-based representations. To validate the proposed A* algorithm under realistic conditions, a real-world case study was conducted using satellite imagery of an urban flood event.

4.2. CA-based grid environment

The grid map was designed in a CA environment, with each cell representing an area of 100 m^2—a scale considered particularly favorable for risk assessment. This size ensures that individuals during evacuation can accurately perceive the environment in a circular region around them [68,8]. Each cell displays water depth and passability, as illustrated in Fig. 7. Obstacles simulate urban areas and evacuation constraints, which account for 20 % of the grid cells, reflecting typical urban built-up area coverage (e.g., buildings, walls, and infrastructure) that often obstruct or redirect movement during flooding events. Water depth levels were categorized as follows: 0.2 m (low risk) to 0.9 m (high risk), with a critical threshold of 0.6 m influencing agent behavior. Optimized A* algorithm's cost function prioritizes safety cost based on DSP rather than explicitly modeling traversal time, as deeper water poses greater hazards (e.g., instability, drowning risk) that exceed traversal time considerations [54]. Higher depths increase safety costs, disfavoring agents from such paths. For example, a cell with a depth of 0.8 m (where walking speed ≈ 0.4 m/s) has a higher safety cost than one with 0.2 m (speed ≈ 1.3 m/s). This dynamic implicitly accounts for traversal time while emphasizing safer, more realistic routing decisions. A total of 20 experiments were conducted using the same setup to evaluate the stability and performance of the improved A* algorithm, measuring path length, critical nodes, explored cells, and execution time. Additionally, comparative

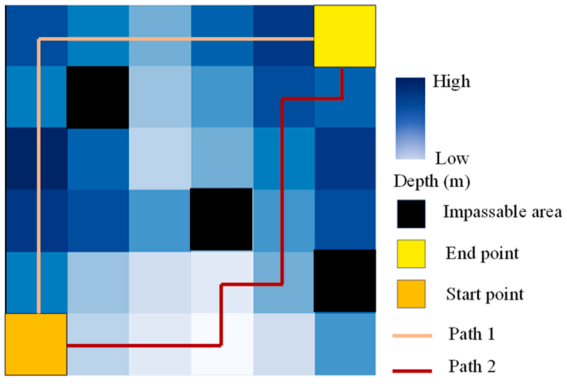


Fig. 6. Illustration of Manhattan distance between two points.

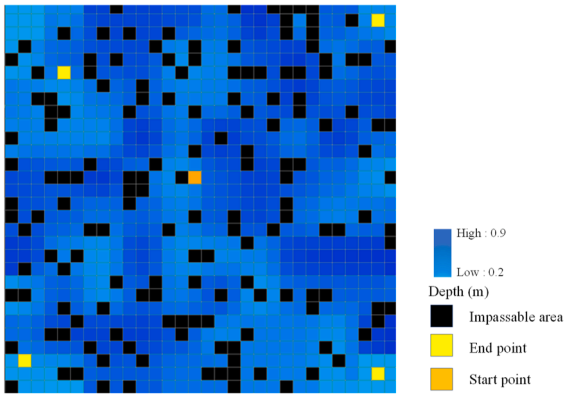


Fig. 7. Grid maps for path planning scenarios.

experiments involving five pathfinding algorithms were performed to assess the enhanced algorithm's effectiveness.

4.2.1. *Proposed A* algorithm simulation*

Path planning simulations evaluated the improved A* algorithm in a 60 × 60 CA grid. Two scenarios were tested: one with a single starting point and multiple endpoints, and a second with multiple starting points. Randomized points, selected to simulate realistic flood conditions, were used. The black blocks are randomly generated obstacles. Results, illustrated in Fig. 8, demonstrate the algorithm's efficiency in finding optimal paths.$(DSP)_{i,t}$ guides the identification of safe rescue paths. In both scenarios, the algorithm highlights the initial paths in gray, while the optimal paths are shown in red, pink, and green.

Table 1 summarizes the results for two scenarios. In Fig. 8a, the algorithm begins with a single start point and generates four initial paths, highlighted in grey, directed to multiple endpoints. These paths are evaluated using a safety cost, calculated by the optimized A* algorithm as a primary heuristic function, and avoiding critical nodes with a depth > 0.6 m. Based on this, the algorithm identifies the optimal path, marked in red, minimizing both safety cost and critical nodes. In Fig. 8b, the algorithm starts with multiple origin points and endpoints, generating six initial paths. Accordingly, with the same depth-sensitive heuristic function, the algorithm identifies three optimal paths corresponding to each start point, highlighted in red, pink, and green. This process highlights the effectiveness of the proposed modified function in prioritizing safer paths. In both scenarios, DSP remained constant despite the need to evacuate through critical cells and cross high-risk areas, confirming its effectiveness in guiding the algorithm to search in low-risk areas and avoid cells considered critical.

Fig. 9(a and b) illustrates changes in water depths along the evacuation path, highlighting the agent's depth sensitivity while avoiding areas exceeding the 0.6 m critical threshold. The algorithm directs the agent to prevent high-depth cells, with short spikes indicating essential decision points when safer paths are unavailable. The study reveals a proportional relationship between safety cost and water depth, as shown in Fig. 9(c and d), becoming more pronounced, emphasizing the importance of depth sensitivity in ensuring agent safety.

Fig. 9(e and f) shows a modeling of the agent's behavior patterns while navigating guided by the improved A-star algorithm. The optimal neighbor cost (blue) reflects the algorithm's ability to evaluate and select the most suitable neighboring cell at each step. This cost includes both safety considerations related to water depth and distance to the exit point. Red markers indicate critical decision points, where agents encountered higher-depth cells, requiring adaptive rerouting. The algorithm's effectiveness is evident in its ability to prioritize low-risk areas while maintaining path efficiency, as the optimal neighbor cost decreases steadily as agents approach

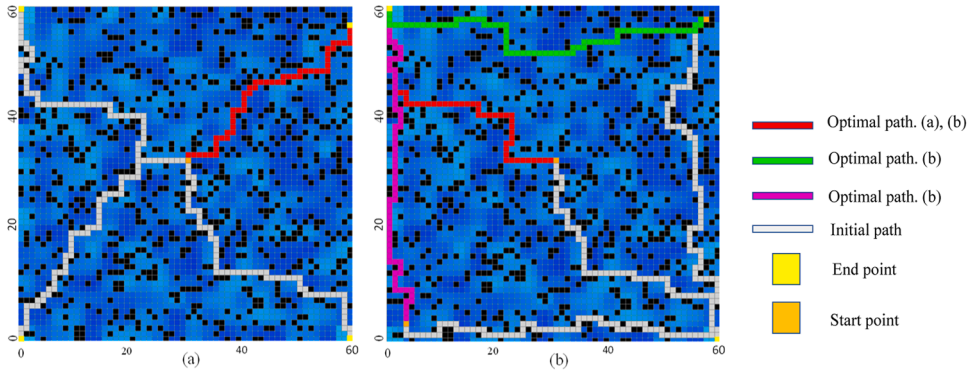


Fig. 8. Path search results using the optimized A* algorithm: (a) Scenario 01 and (b) Scenario 02.

Table 1
Scenarios a and b metrics in a grid environment.

Scenarios	Path	Path length	Safety cost	Critical nodes	Explored cells	Execution time (s)	Depth sensitivity
(a)	Red	53	266.68	8	134	1.47	0.56
(b)	Pink	71	331.15	20	165	2.81	0.52
	Red	63	276.13	9	152	1.81	0.49
	Green	73	387.80	24	253	3.41	0.59

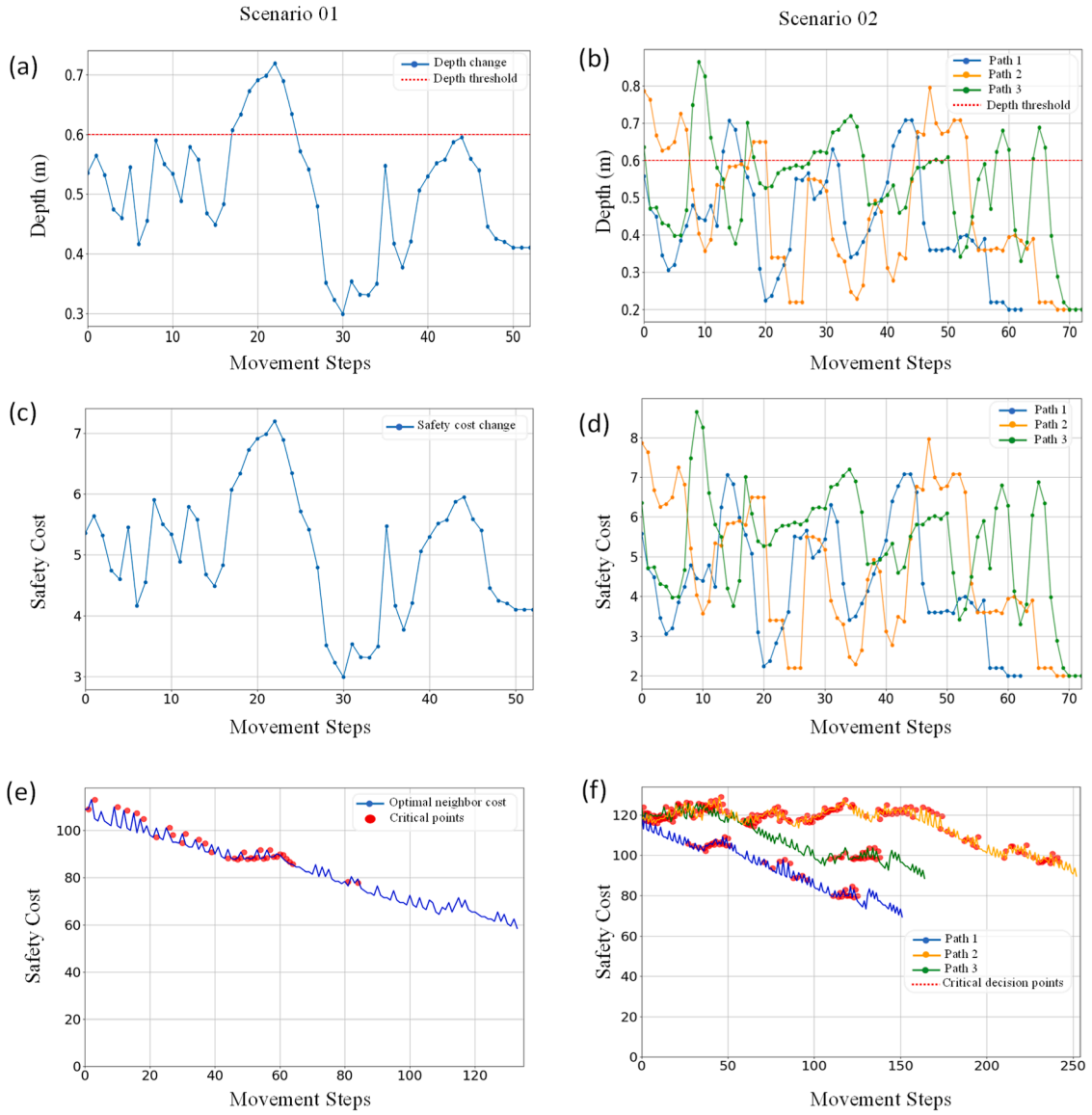


Fig. 9. Path search analysis in CA grid environment for two scenarios: (a–b) depth variation, (c–d) safety cost change, and (e–f) cost evolution with critical decision point transitions.

the exit.

4.2.2. Simulation comparison between different algorithms

The performance of the improved A* algorithm was rigorously evaluated against five established path planning algorithms on a 100 × 100 cell grid (Fig. 10). Path generation followed the predefined criteria outlined in Table 2, with the target point positioned at the lowest water depth to ensure alignment with safety objectives. To facilitate a consistent comparison, the initial (0, 100) and target

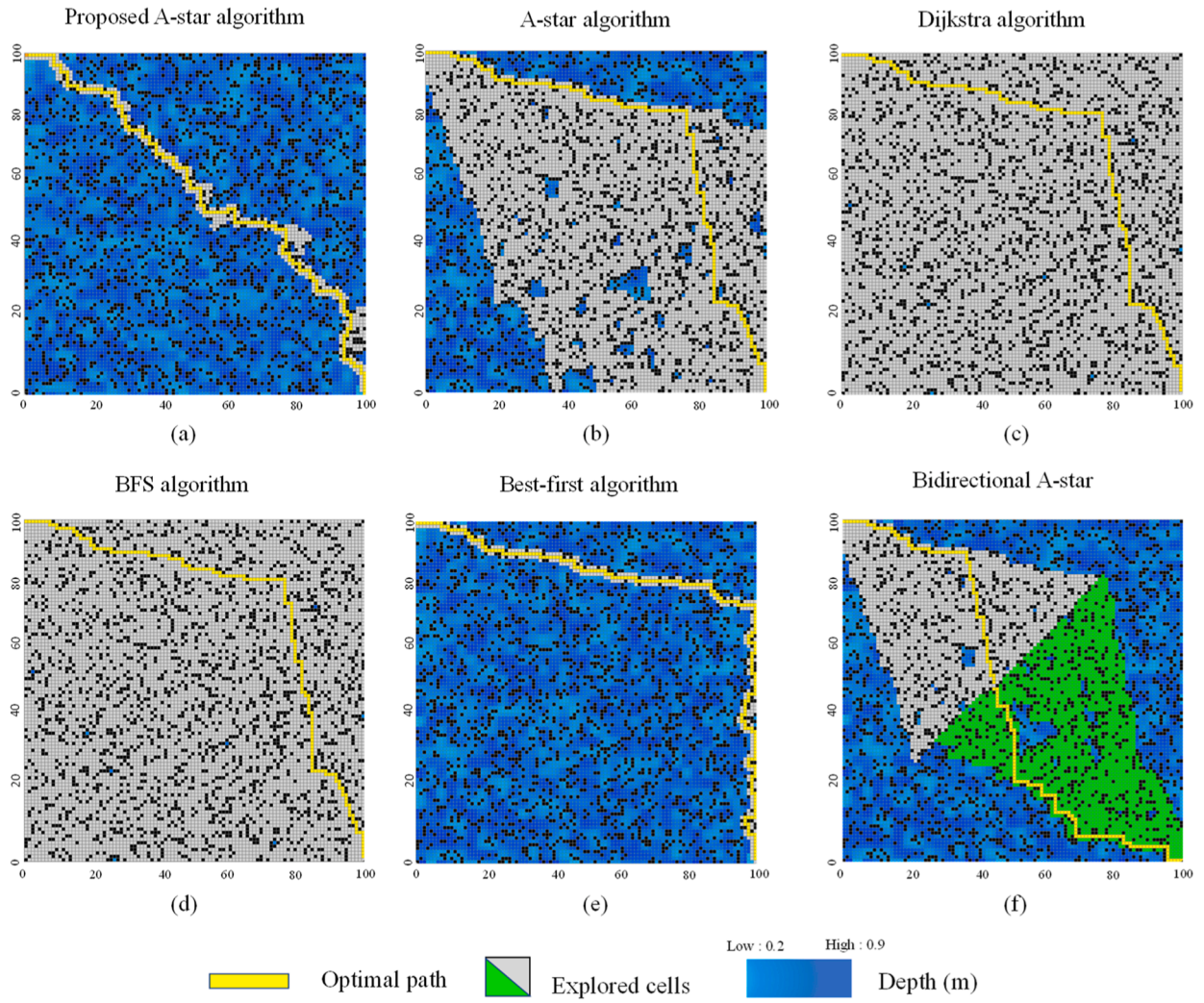


Fig. 10. Comparative path search results generated by different algorithms: (a) proposed A* algorithm, (b) original A* algorithm, (c) Dijkstra algorithm, (d) BFS algorithm, (e) best-first algorithm, and (f) bidirectional A*.

Table 2
Simulation results of six algorithms on a 100 × 100 grid map.

Indicators	Original A*	Dijkstra's	BFS	Best-first	Bidirectional	Optimized A*
Execution time (s)	191.88	319.69	366.41	18.53	158.41	22.89
Path length	199	198	198	224	107	206
Explored cells	5502	7972	7972	455	4159	547
Critical nodes (>0.6 m)	70	172	70	89	49	39

(100, 0) points remained fixed across all six test maps. The algorithm-generated path is depicted in yellow, while explored cells are marked in gray and green. Results demonstrate the enhanced algorithm's superior performance in balancing safety prioritization and computational efficiency. Compared to Dijkstra's algorithm, the optimized A* reduced execution time by 88.06 %, while improvements of 93.76 % and 87.65 % were observed against BFS and the standard A* algorithm, respectively. Additionally, the number of explored cells decreased by 93.13 % relative to BFS, 90.06 % compared to the original A, and 93.13 % versus Dijkstra's algorithm. Notably, critical nodes—defined as cells with water depths exceeding 0.6 m—were reduced by 44.29 % and 77.33 % compared to the original A and Dijkstra's, respectively, and by 56.18 % relative to BFS. While the best-first search exhibited the shortest execution time, it lacked safety considerations. The bidirectional algorithm achieved the minimal path length by concurrently searching from both start and end points, thereby optimizing distance coverage. The improved A* slightly increased path length by 3.52 % compared to the original A*, reflecting its prioritization of safer routes over minimal distance. Ultimately, the enhanced A* algorithm strikes an optimal

balance between computational efficiency and safety, effectively minimizing critical node exposure, reducing computational overhead, and leveraging depth sensitivity to guide agents along low-risk evacuation paths.

4.3. Realistic case study

4.3.1. Flood depth pixel map construction

Following the modeling approach detailed in Section 3.1, a realistic case study was developed using flood data from Hurricane Harvey in Houston, Texas (August 17 to September 3, 2017). Depth level data were obtained from HydroShare (https://www.hydroshare.org/resource/165e2c3e335d40949dbf501c97827837/), and corresponding aerial imagery was sourced from Woolpert Maps (https://maps.woolpert.com/harvey/). The geospatial datasets were processed in QGIS to create a simulation-ready environment. The floodwater data were reprojected to a Universal Transverse Mercator (UTM) coordinate system and resampled to a uniform resolution of 10 × 10 m per pixel. This rasterization ensures each pixel corresponds to a CA grid cell, representing a discrete spatial unit with a water depth value. As depicted in Fig. 11, this rasterized pixel map provides a grid-aligned CA framework for agent positioning and flood condition assessment. In the realistic study, floodwater is treated as still based on peak inundation data, aligning to simulate worst-case evacuation scenarios. This pixel grid environment forms the foundation for simulation in subsequent sections.

4.3.2. Environment-based scenarios

The optimized A* algorithm's performance was evaluated under different spatial constraints using three realistic evacuation scenarios based on the flood raster map introduced in Section 4.3.1. To streamline pathfinding efficiency and focus on evacuation behavior in static environments, the simulation treated only buildings as impassable obstacles while excluding dynamic obstructions such as vehicles and debris.

The simulation environment was developed in two phases: map generation and path planning. Initially, aerial images of flooded areas were combined with depth data to create a water depth map. Building and road network layers were converted to raster formats and merged to produce a comprehensive map featuring water depth, building distribution, and road networks. In the second phase, this final map was used as input for an enhanced A* algorithm to generate optimized evacuation and rescue paths across three scenarios, testing robustness and efficiency. Start points were strategically selected to create complex pathfinding challenges, while endpoints were chosen based on low water depth. The scenarios included (1) path planning constrained to the road network (Fig. 12a), (2) path planning over open flooded areas with fixed obstacle avoidance (Fig. 12b), and (3) path planning for multiple origins navigating to a single destination (Fig. 12c), where multiple points represent a group of agents simulating crowd movement. For consistency, all agents were assumed to be adults with equal physical characteristics and decision-making thresholds. Pathfinding computations were conducted on a 10 × 10-meter resolution flood depth raster, ensuring alignment with the CA model structure. The

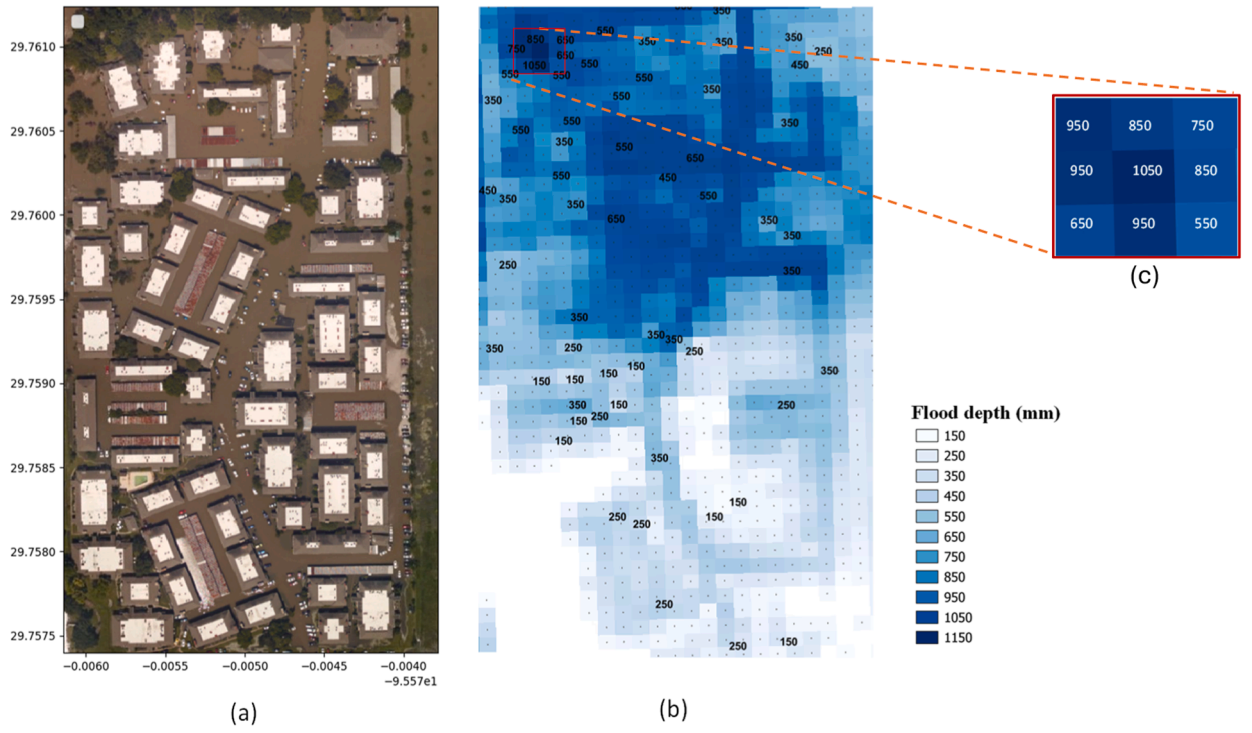


Fig. 11. Pixel-based CA environment construction: (a) aerial image of flooded area, (b) rasterized depth map at 10 × 10 m resolution, and (c) zoomed-in grid showing a cell with its eight surrounding neighbors and depth values.

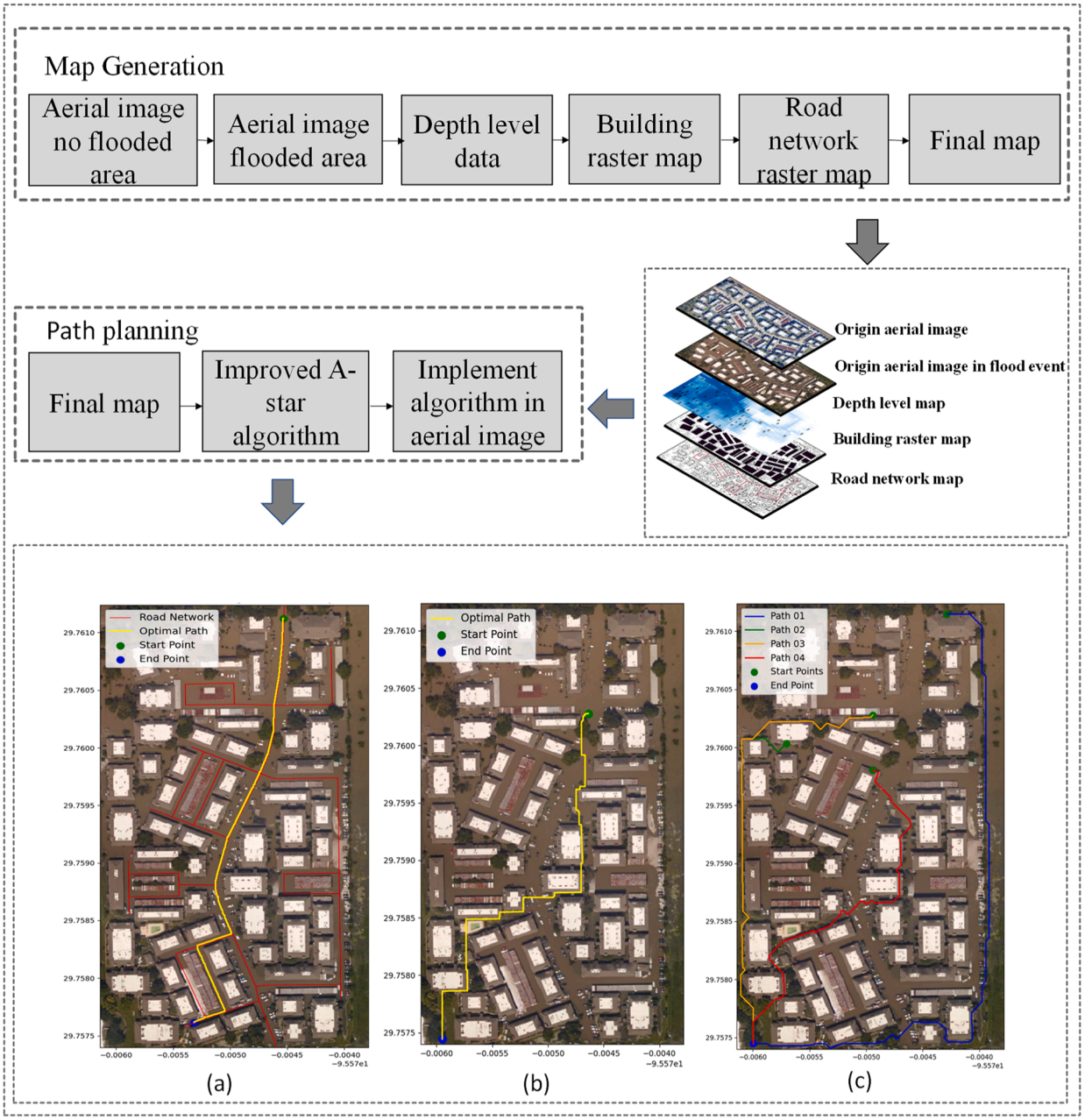


Fig. 12. System workflow for path planning implementation across three aerial image scenarios (a, b, and c representing Scenario 1, 2, and 3, respectively).

simulation results were visualized over the aerial image for clarity.

Fig. 13 illustrates the comparative results from three scenarios. In Scenario 01, the agent is restricted to the road network, limiting decision-making flexibility. Consequently, it follows the shortest path but encounters multiple critical depth areas (>600 mm), failing to avoid risks due to limited mobility. Scenario 02 allows the agent to evacuate freely while avoiding obstacles, enabling the enhanced algorithm with DSP to dynamically assess surrounding depth values and prioritize safer paths. The heuristic penalizes movement through deeper flood zones, guiding the agent toward areas with lower perceived risk, evident in the rapid decline of depth levels over time. Isolated spikes in depth indicate critical decision points where agents trade safety for evacuation continuity, reflecting real-world urgency. Scenario 03 simulates multi-agent decision-making, with each agent selecting independent paths based on interactions with depth levels and perceived risks. These paths demonstrate adaptive behavior, as agents actively avoid highly inundated areas.

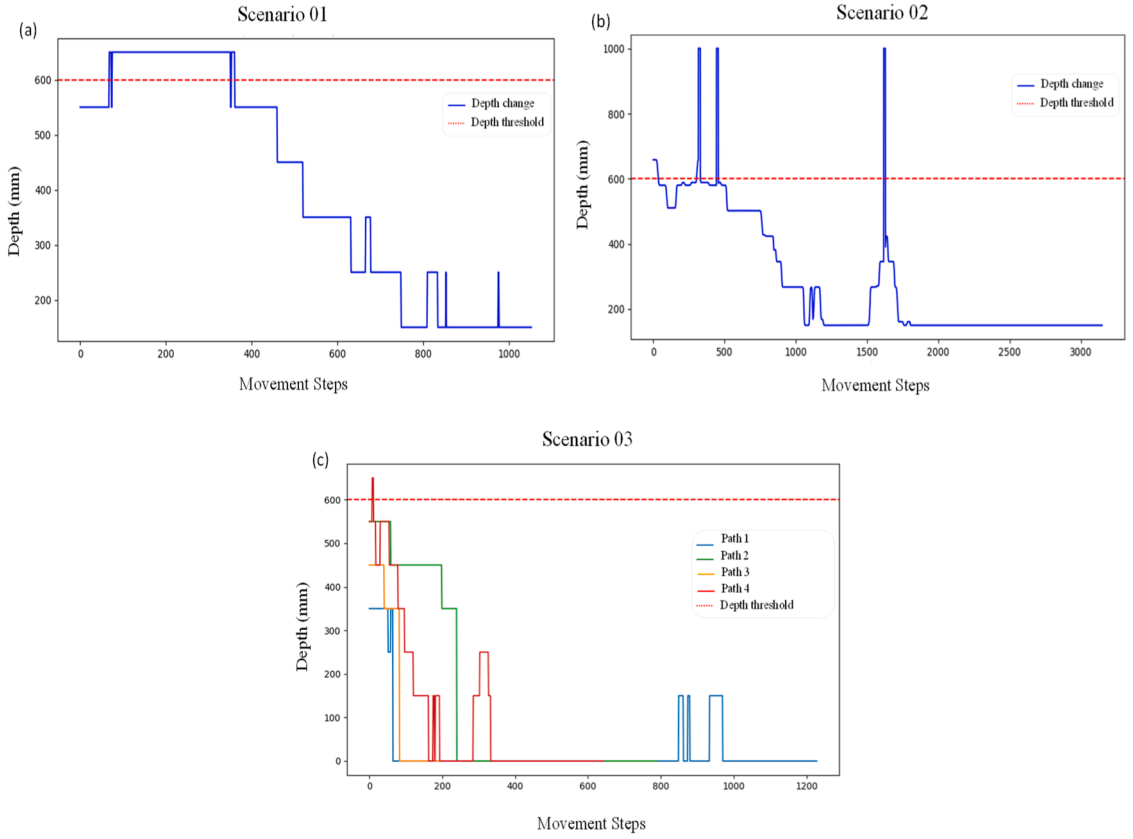


Fig. 13. Path search analysis in a realistic study environment for three evacuation scenarios: (a) constrained road-based evacuation, (b) evacuation avoiding obstacles and high-depth areas, and (c) multi-agent path planning.

4.3.3. Agent height-based scenarios

To further investigate the impact of physical characteristics on evacuation behavior during floods, a supplementary modeling analysis was conducted with a specific focus on agent height as a determining factor. The study simulated agents of varying heights (1.2 m, 1.7 m, and 1.9 m) within identical flood conditions while maintaining consistent environmental parameters to enable cross-scenario comparisons. Each agent initiated from a different starting point, while all agents evacuated at a single endpoint set in a low-depth area considered a safe shelter. The path length between the starting and ending points was purposefully extended to showcase a worst-case scenario in evacuation path planning. Each agent's flood risk perception was dynamically calibrated according to water immersion levels relative to their height, following established immersion thresholds [14]. The risk classification system categorized flood depths into four distinct levels: minimal risk (depth $\leq$ 10 % of agent height, approximating ankle-level water), moderate risk (10 % $<$ depth $\leq$ 30 % of height, reaching knee level), high risk (30 % $<$ depth $\leq$ 55 % of height, between knee and waist), and critical risk/immobilization (depth $>$ 55 % of height, above waist level). During pathfinding computations, agents continuously evaluated their surroundings, designating cells exceeding critical immersion thresholds as non-navigable. The enhanced A* algorithm prioritized safety over minimal distance in its path generation, producing height-specific evacuation routes. The simulation outcomes for all three agent height variations across multiple scenarios are comprehensively illustrated in Fig. 14.

Fig. 15 illustrates the results of three agent-height-based evacuation scenarios. In evacuation scenario 01, the 1.2-meter agent adheres to a constrained path, frequently encountering critical water depths relative to its height. Dynamic risk perception profoundly impacts evacuation behavior as the agent continuously navigates high-risk areas, reflecting heightened immersion exposure. Scenario 02 demonstrates improved adaptability with the 1.7-meter agent, which avoids critical areas when possible and selectively traverses moderate-risk cells, guided by the DSP-based algorithm. The gradual depth reduction along its path underscores safer evacuation decisions informed by real-time environmental assessment. Scenario 03 features the most resilient 1.9-meter agent, which traverses deeper areas with minimal risk due to its lower immersion sensitivity. The path remains within minimal to moderate risk thresholds, capturing adaptive behavior with minimal exposure to critical areas. Across all scenarios, sporadic depth increases mark critical decision points where agents accept temporary risks to preserve evacuation continuity, emphasizing the behavioral realism captured by the depth-sensitive path planning model.



Fig. 14. Height-dependent evacuation path planning for agents of varying statures (1.2 m, 1.7 m, and 1.9 m).

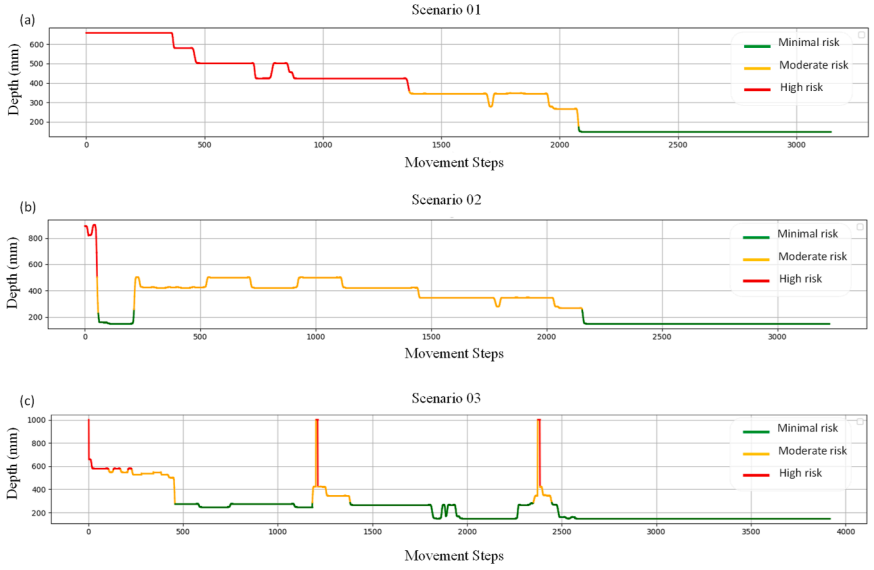


Fig. 15. Path search analysis in height-based evacuation simulation for three scenarios: (a) 1.2 m agent, (b) 1.7 m agent, and (c) 1.9 m agent under identical flood conditions.

5. Discussion

5.1. Optimized A* algorithm for urban flood evacuation

The optimized A* algorithm addresses critical challenges in urban flood evacuation path planning using DSP and safety-cost heuristics. These advancements overcome limitations in traditional algorithms by prioritizing safety alongside efficiency, particularly in uncertain and complex environments. By continuously evaluating real-time water depths and agent behavior within a CA framework, the algorithm effectively mitigates safety risks and enhances evacuation outcomes. Key findings include:

- DSP integration allows agents to make realistic, risk- perceived evacuation decisions in flood conditions.
- Optimized A* algorithm outperforms traditional methods in execution time, safety-cost reduction, and adaptability to urban environments.
- CA-based grid environment allows adaptive modeling of urban conditions and simplifies the simulation between idealized and real-world scenarios.
- Supporting a behavioral perspective on flood evacuation, showing how environmental risk perception effects path choice and decision making.
- The proposed approach using ABM can be integrated with sensor inputs, and immersive technologies for future smart evacuation planning systems.

Building on advanced studies of stochastic and heterogeneous conditions [69], this research demonstrates superior adaptability across diverse flood scenarios while achieving higher computational efficiency and decision-making accuracy than traditional methods. The proposed approach enriches path-planning research by integrating behavioral dynamics with spatial risk analysis, offering a robust solution to the ultra-complex challenges of urban flood rescue. Experimental results validate the algorithm's ability to balance safety and efficiency, establishing a foundation for advancing flood evacuation modeling under extreme natural hazards.

The findings align with microscopic approaches that represent individual responses to floodwater conditions, including ABMs [70], and complement force-based models to evaluate pedestrian behavior and movement velocities [47]. This work highlights the importance of human behavioral features in evacuation, emphasizing the need to integrate risk assessments where the optimal path may involve traversing high-depth areas as the least harmful option [14,71]. Furthermore, agent height simulations indicate how physical characteristics affect evacuation behavior: shorter individuals exhibit redirected paths due to lower immersion thresholds, while taller agents navigate deeper floods more directly. These insights highlight the necessity of incorporating DSP with human variability for realistic modeling.

To enhance practical applications, the study advocates integrating communication strategies, such as mobile applications and social media, to reduce evacuation response times and path choice [72]. Machine learning techniques can further analyze the situational factors influencing behavior [73], while structured databases within the flood context provide a reliable basis for identifying key decision drivers. The depth-sensitive inference method developed here models human responses to water depth, enabling optimal path identification. Future research could combine these quantitative insights with immersive technologies like virtual reality (VR) or augmented reality (AR) to refine behavioral representation in dynamic flood scenarios [24,74]. Such advancements will strengthen risk mitigation strategies and evacuation planning, ultimately saving lives in flood-prone urban environments.

5.2. Limitations and future research directions

While this study provides helpful perspectives on flood evacuation pathfinding, several limitations remain and offer clear directions for future research. First, the reliance on a fixed-resolution grid (10×10 m), although computationally efficient and compatible, introduces spatial constraints. Buildings with sides smaller than 10 m may be underrepresented, potentially restricting the accuracy of obstacle modeling and evacuation path simulation in densely urban areas. Moreover, the current approach does not incorporate plano-altimetric variation, particularly variations in terrain elevation and slope, which significantly affect flood dynamics and pedestrian movement. Future work should consider using a multi-level grid resolution, wherein preliminary evacuation mapping and DSP assessments are performed at a more precise grid level, and critical risk areas are subsequently analyzed at a finer spatial resolution. This recommendation would allow for more detailed modeling of human-scale interactions, including realistic obstacle dealing and risky path analysis. Furthermore, the assumption of static flood conditions restricts the model's applicability to real-world scenarios where dynamically rising or receding water occurs. The exclusion of moving obstacles (e.g., vehicles, debris, or crowds) further limits the behavioral realism of evacuation scenarios.

From a behavioral perspective, the model does not explicitly account for key factors influencing human responses to floods, such as physical ability, cultural background, risk perception, and environmental awareness. Moreover, while the optimization of individual paths is addressed, collective evacuation strategies or community-level interactions —essential for urban resilience—remain unexplored. Future work should investigate how behavioral modeling can generate adaptive evacuation maps that respond to real-time environmental changes and evolving human decisions. Potential solutions include integrating agent-based simulation with real-time data streams (e.g., IoT sensors, mobile apps) and machine learning-based models to dynamically update risk zones and evacuation priorities based on perceived safety. Addressing these gaps through dynamic data assimilation, high-resolution modeling, and refined behavioral frameworks will enhance the realism and policy practicality of evacuation planning tools.

6. Conclusion

This research proposed an optimized A* pathfinding algorithm for flood evacuation that incorporates DSP and safety-cost-based heuristics within a CA and ABM framework. The improvements allow agents to prioritize safer routes under flood risk, enhancing decision-making efficiency while avoiding high-risk zones. Experimental results, across both simulated and real-world scenarios, demonstrate the algorithm's superior performance over traditional methods in minimizing search complexity (reduction of up to 93.13 % in explored cells and 88.06 % in execution time compared to Dijkstra's algorithm) and improving evacuation outcomes. Its adaptability to diverse flood conditions, including structured roads, urban obstacles, and open areas, confirms its practical use in emergency planning.

Overall, this study contributes a scalable, behavior-driven path planning model that is highly applicable to real-world emergency evacuation scenarios during floods. The optimized algorithm can be effectively applied across various urban environments, including structured road networks, obstructed zones, and open flooded areas, offering a robust tool for emergency response planning. Its integration with sensing technologies, such as drones, flood monitoring systems, and IoT-based infrastructure, enhances its utility in real-time decision support. Beyond human evacuation, the model also holds potential for logistical coordination, including the distribution of critical resources during disaster events. While the model expands behavioral realism with spatial risk analysis, the proposed optimization offers a robust framework as a practical solution for improving spatial resilience and supporting flood risk management at both strategic and operational levels.

CRediT authorship contribution statement

Housseyn Chebika: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration. **Guoqiang Shen:** Conceptualization, Methodology, Formal analysis, Resources, Data curation, Writing – review & editing, Visualization, Supervision, Project administration. **Haoying Han:** Writing – review & editing, Supervision. **Mahmoud Mabrouk:** Writing – review & editing, Supervision. **Brahim Nouibat:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

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Further reading

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